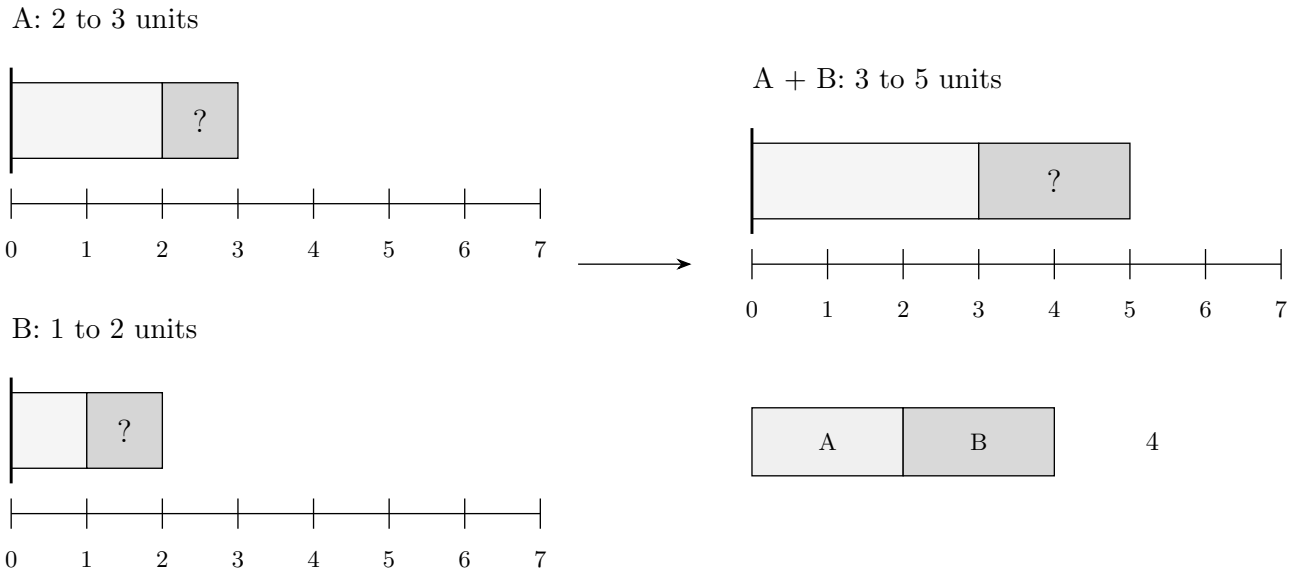
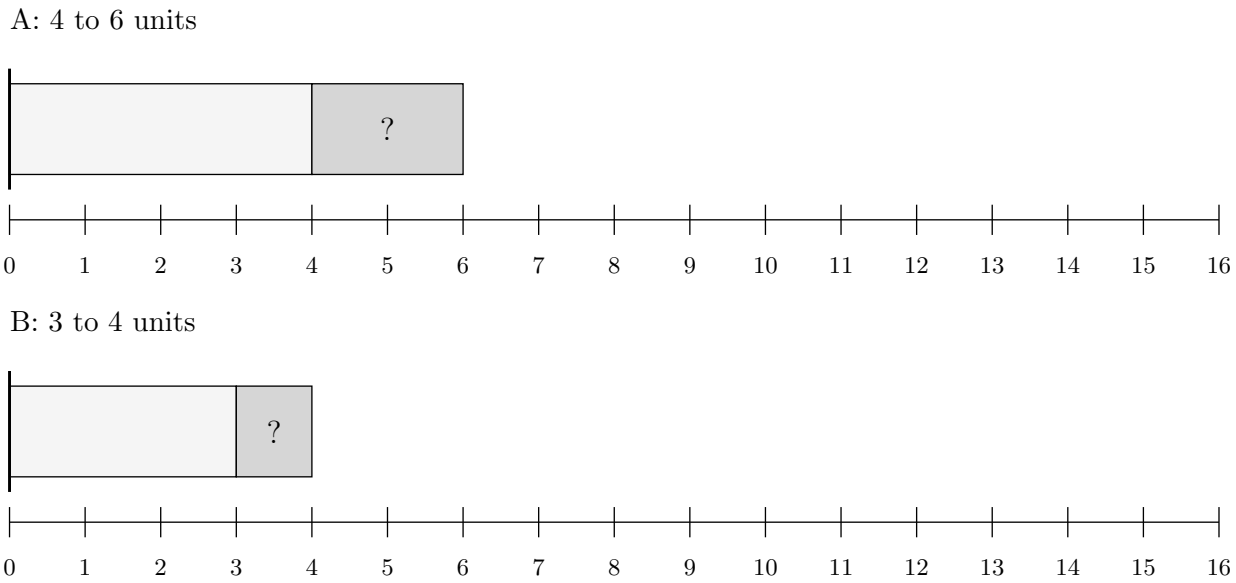
A hidden end can be anywhere in its gray band, including the edges. Each trial may use new lengths. Join strips end to end from 0, with no gap or overlap.

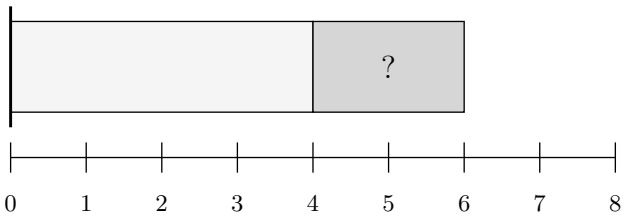


Problem 1: Make the shortest and longest joins with these strips. Can you make a join ending exactly at 8?

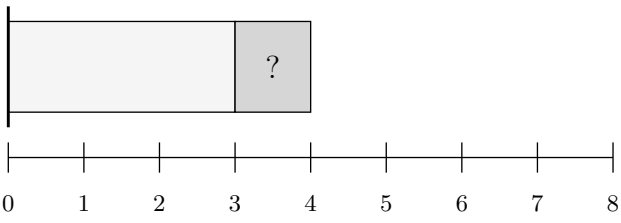


Problem 2: Try A with B, C with D, and E with F; which pairs always reach 8? Let a friend try to defeat your choices.

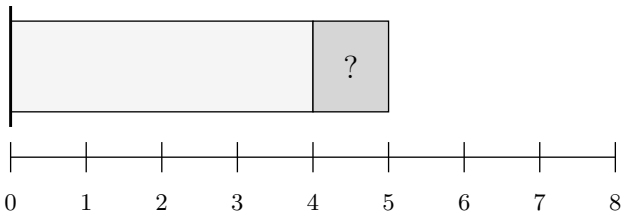
A: 4 to 6 units



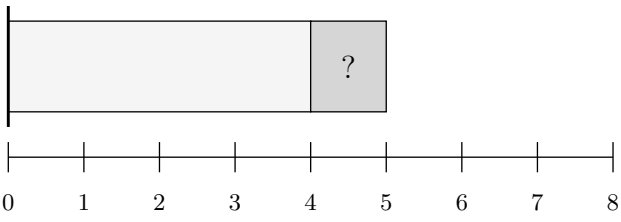
B: 3 to 4 units



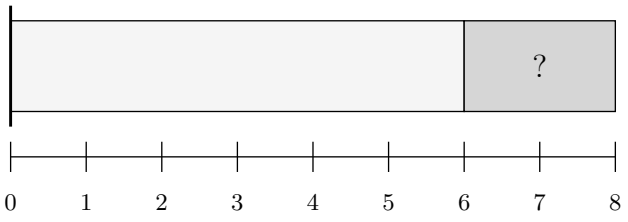
C: 4 to 5 units



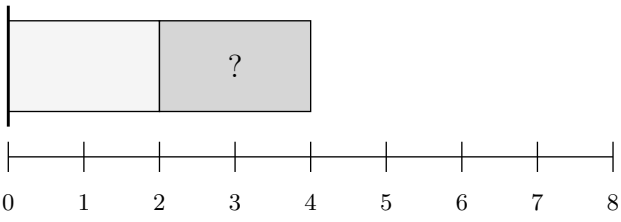
D: 4 to 5 units



E: 6 to 8 units



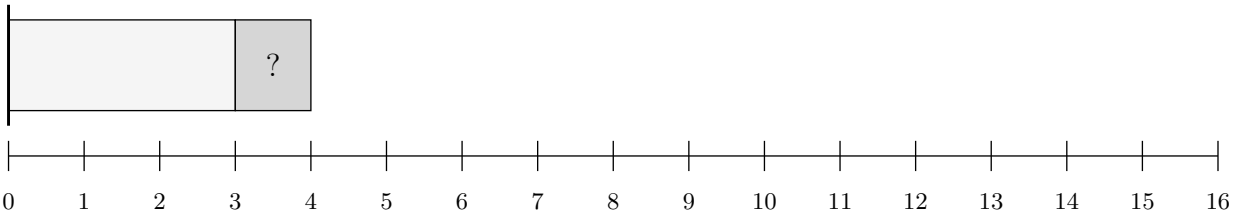
F: 2 to 4 units



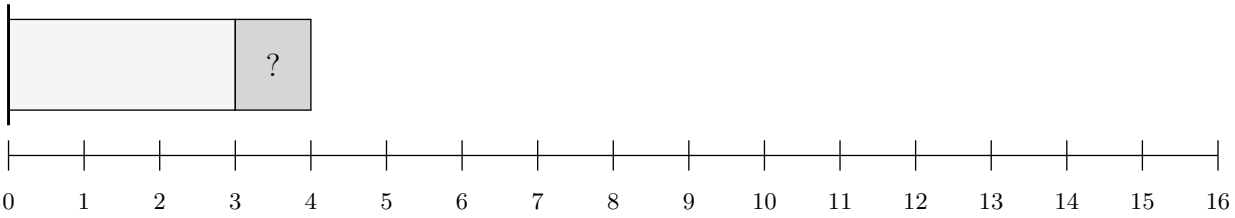
Reaching 8 includes ending at 8.

Problem 3: Pick a card for B so the join always reaches 7 and fits within 9. Try to break your choice with different allowed strips.

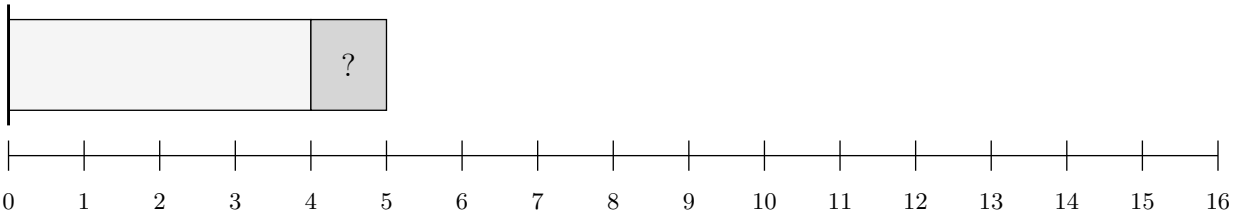
A: 3 to 4 units



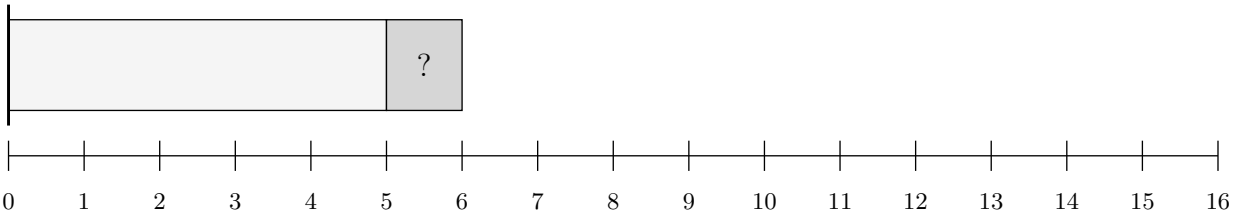
B: 3 to 4 units



B: 4 to 5 units



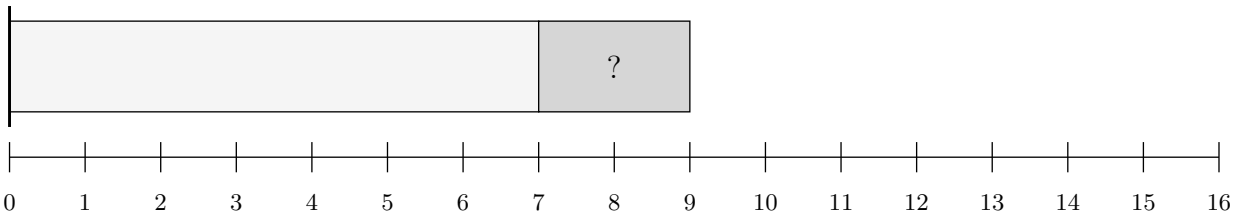
B: 5 to 6 units



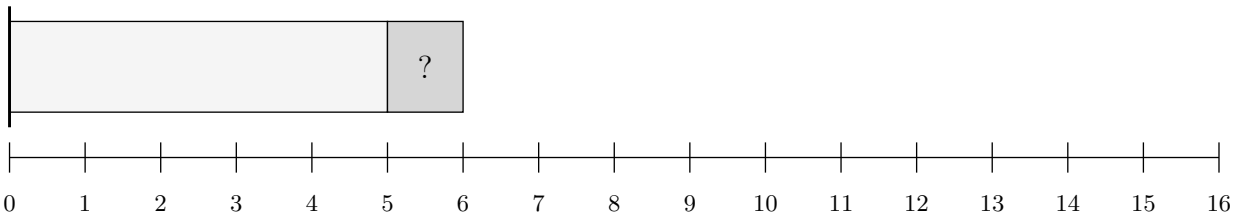
Fitting within 9 includes ending at 9.

Problem 4: Put both starts at 0. Make the biggest gap between the ends, then the smallest.

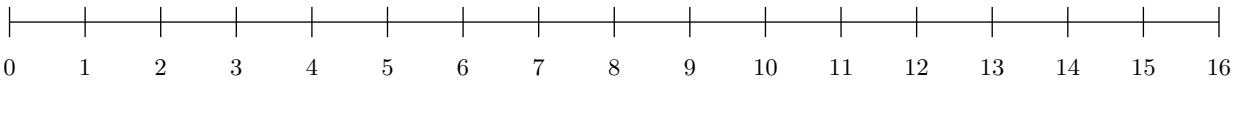
long strip: 7 to 9 units

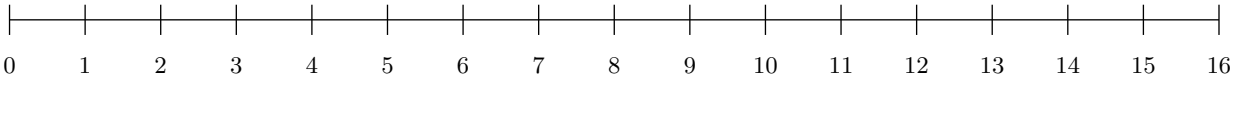


short strip: 5 to 6 units



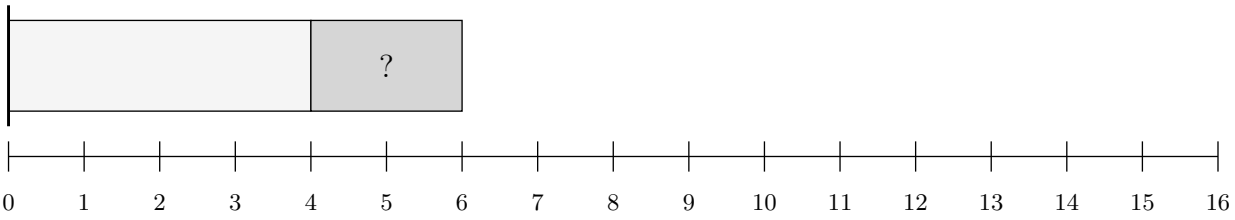
Problem 5: Using these two strips, make a gap of 3 in two different ways.



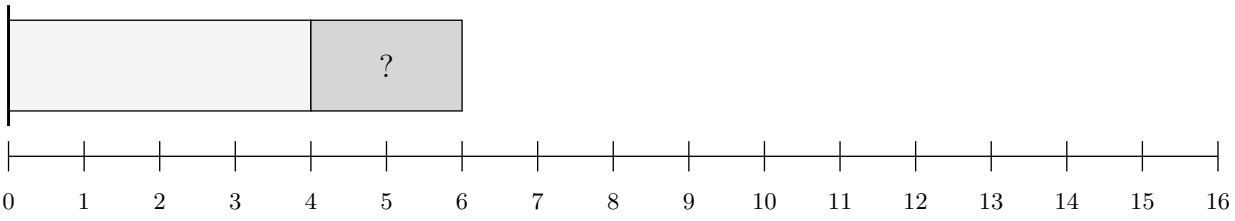


Problem 6: Find how small and how big each end gap can be. Start at 0. Both A's in the top picture are the same length. Each try may use new lengths.

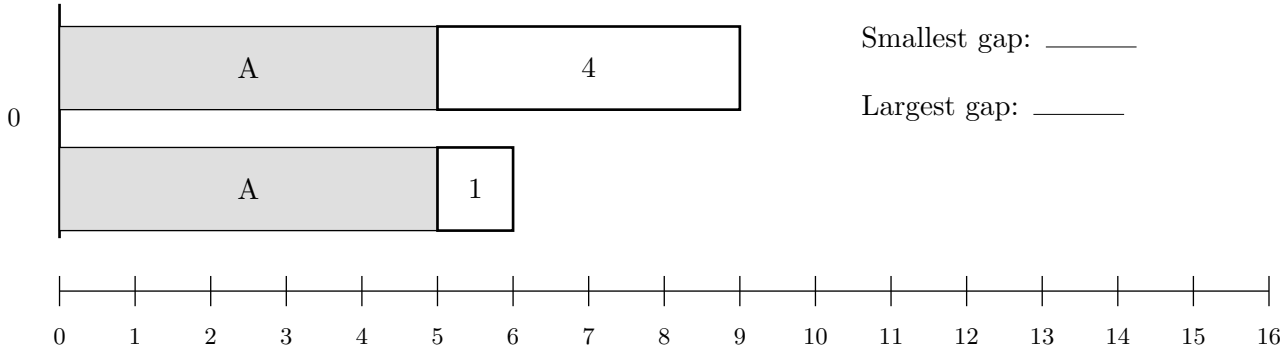
A: 4 to 6 units



B: 4 to 6 units

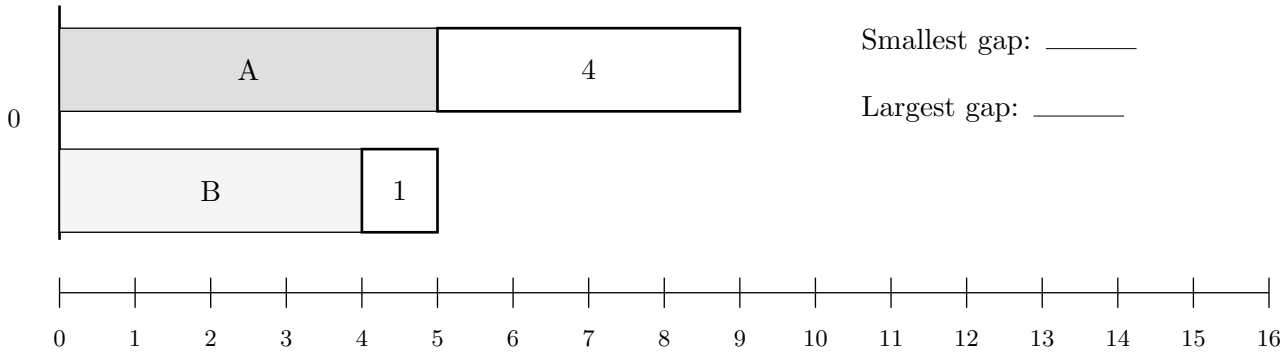


The white pieces are exactly 4 and 1 units long.



Smallest gap: _____

Largest gap: _____

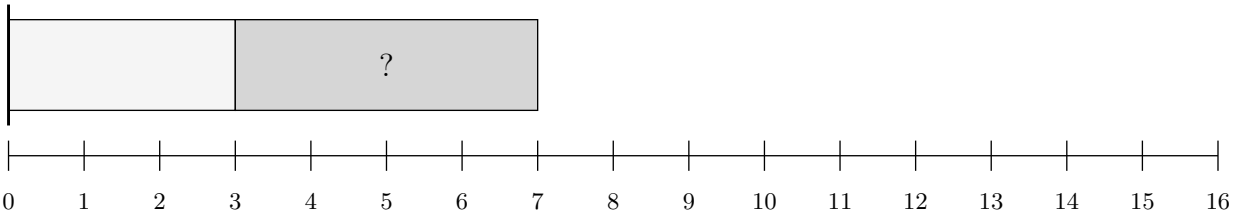


Smallest gap: _____

Largest gap: _____

Problem 7: A and B must fill 9 units exactly. Find every pair of whole-unit lengths that fits both cards.

A: 3 to 7 units



B: 3 to 5 units

